

WASP-12b

R-1368

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_200219 · created 2026-08-01T00:26:07+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458898.66706 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.046
Transit depth (Rp/Rs)^2	2.33 +/- 1.4 [%]
Orbital Inclination (inc)	85.82 +/- 2.95
Airmass coefficient 1 (a1)	0.052 +/- 0.039
Airmass coefficient 2 (a2)	2.99 +/- 1.67
Scatter in the residuals of the lightcurve fit is	89.34 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	4.25
Transit Duration (day)	0.131 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.046 vs 0.1178 (deviation 0.76σ)
Duration fit vs published	0.131 d vs 0.125 d (deviation 1.12σ)
Mid-time O–C	0.7 min
Depth SNR	3.3
Residual scatter	89.34 %

Light curve

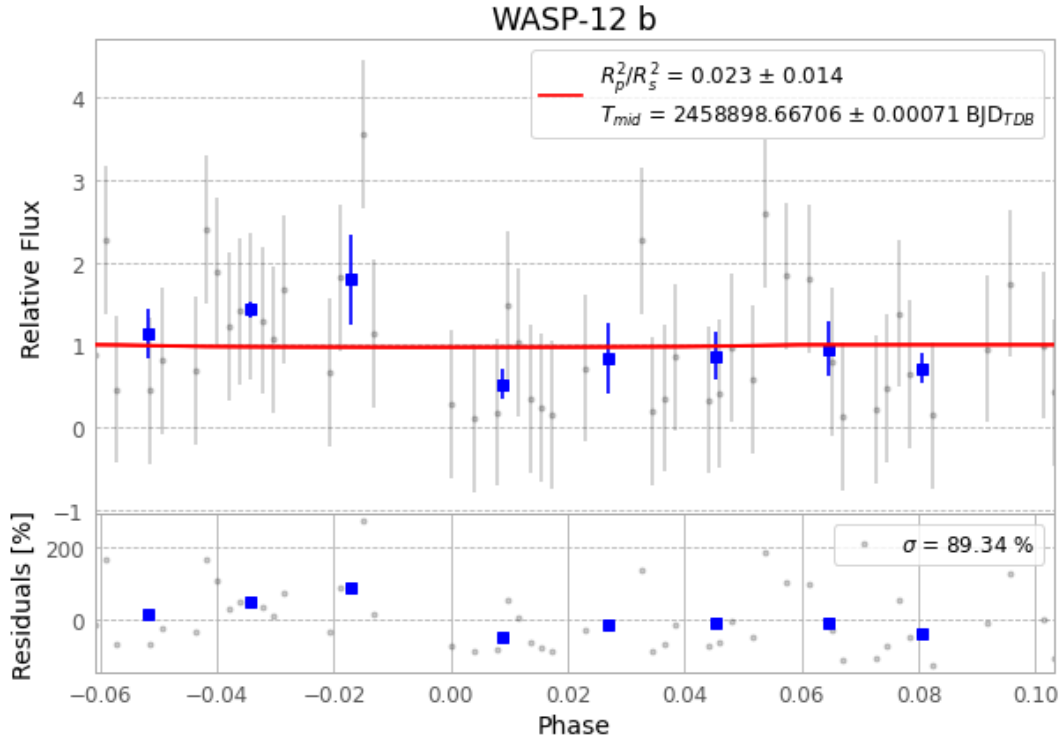


Figure 1 — FinalLightCurve_WASP-12 b_2020-02-18.png (SHA-256 c44cd0e50f1d716e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 343], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0af0bbf6b2e6501a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), WASP-12200219021812.FITS → WASP-12200219064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200219.log (d22ef5434a7d...), FinalLightCurve_WASP-12 b_2020-02-18.png (c44cd0e50f1d...), FinalLightCurve_WASP-12 b_2020-02-18.csv (2f092482315f...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:26Z.